DIC05
Miller Capacitance

Specification-based laboratory report

17 October 2026

English edition
HSPICE S-2021.09 | Model library supplied separately

1 Objectives

Follow Lab 5 (PDF pages 21–23) to observe Miller coupling at the second stage of a three-inverter chain and quantify rail excursions and loading.

2 Specification mapping

Handout requirement (PDF page)	Implementation	Verification and scope
Step 1: three inverters and stage-2 glitches (21–23)	01_miller, case 0	Simulated with native parasitics
Step 2: add and vary C_c (22–23)	Cases 1–3: 2/5/10 fF	Includes the specified 5 fF
Question: increase output load (23)	Cases 4–5: add 10/40 fF	Compare at fixed 5 fF coupling

3 Circuit and design derivation

All three inverters use $W_n/W_p/L = 0.6/1.8/0.25\,\mu$m. C_c spans the second-stage input and output. Stage 3 remains a gate load, while additional C_L connects to ground. The qualitative relation $\Delta V_o \approx C_c \Delta V_i/(C_c + C_{\text{out}})$ explains charge division; nonlinear MOS simulation supplies the actual values. Scattered source cases are replaced by comparable parameter cases with explicit peak and edge measurements.

4 Simulation setup

This teaching rewrite uses a separately supplied model at 1.0 V and 25 °C. The specified topology, loads and comparative dimensions are retained. The handout's 2.5 V process results are not acceptance values for this model. The chosen 1 V is a simulation condition, not a verified device rating. MOS bulk terminals are explicitly connected to the corresponding supply rails.

HSPICE S-2021.09 uses RUNLVL=6, retaining waveforms, printed data and native measurements. Figures are redrawn from actual data; CSV files are in each Lab's results directory. Successful solution does not establish process reliability. Warnings from encrypted model blocks remain suppressed; the existing diagnostic did not expose actionable further details, so the warnings are not declared harmless and complete model-limit compliance is not claimed.

Input edges are 0.1 ns, with a 1 ns initial delay, 2 ns high interval and 5 ns period. Transient duration is 10 ns with 1 ps printing. Peaks are measured over 0.8–5 ns; rise/fall use 10–90%/90–10% crossings.

5 Results and specification checks

All six cases concluded normally without errors or failed measurements. C_L is additional to the stage-3 gate load.

C_c (fF)	C_L (fF)	Over (mV)	Under (mV)	t_r (ps)	t_f (ps)
0	0	123.09	83.47	133.70	76.70
2	0	154.03	130.47	155.08	99.82
5	0	188.80	182.72	189.49	133.95
10	0	227.81	237.61	250.08	190.65
5	10	143.98	142.61	279.78	178.15
5	40	80.57	80.85	558.95	318.36

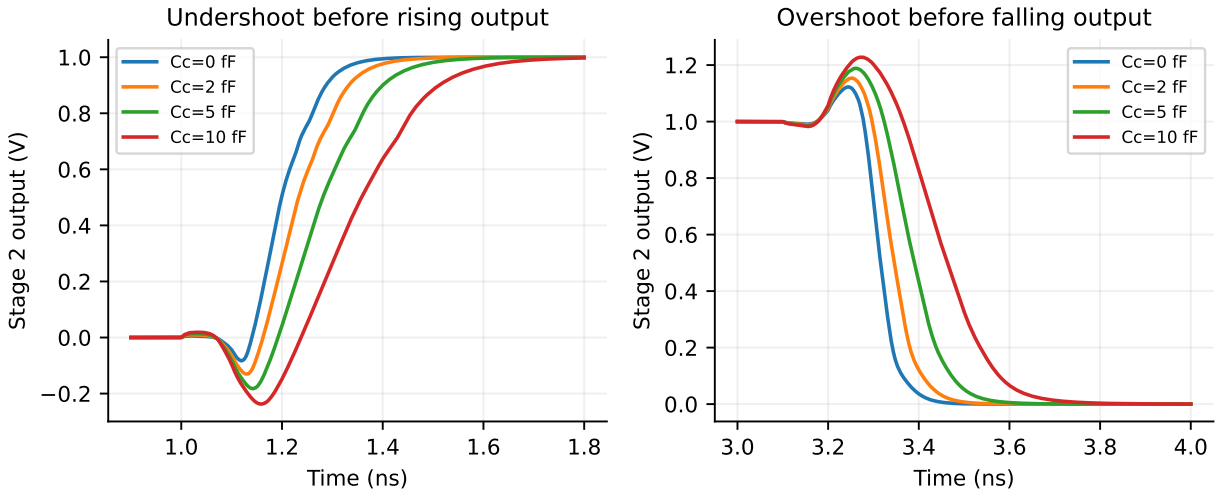


Figure 1: Transitions and rail excursions at stage 2; redrawn from this run's HSPICE data.

6 Discussion

Increasing C_c from 0 to 10 fF raises overshoot from 123.09 to 227.81 mV and undershoot from 83.47 to 237.61 mV. At 5 fF coupling, adding 40 fF load reduces the peaks to about 80.6/80.9 mV while increasing rise/fall times to 559/318 ps. Zero added coupling does not remove intrinsic Miller capacitance. Rail excursions are simulation results; they do not establish hardware reliability or immunity of every receiver.

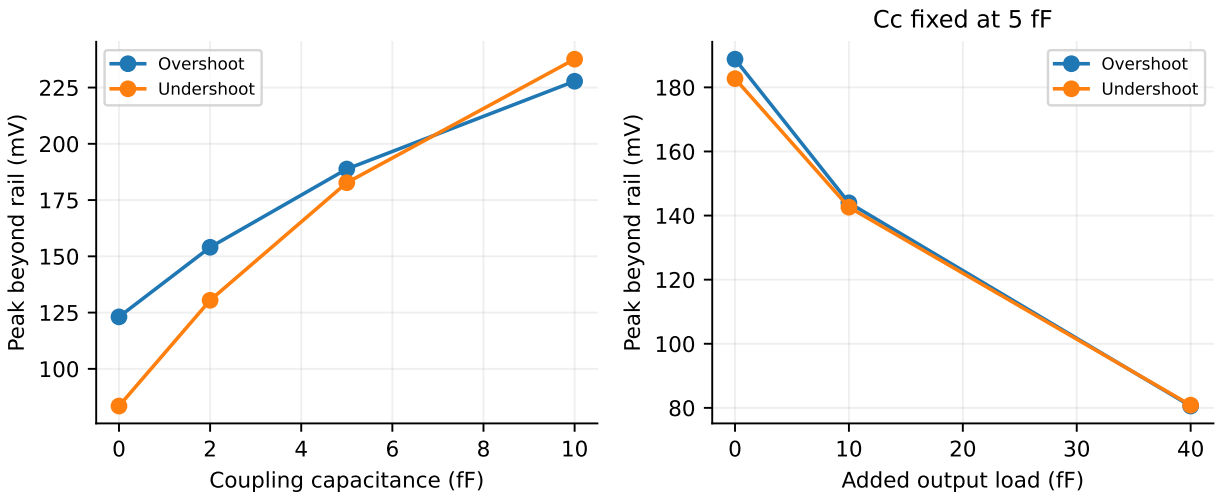


Figure 2: Effect of coupling capacitance and additional output load; redrawn from this run's HSPICE data.

7 Conclusion

Both experimental steps and the loading question are covered by actual data. Results support capacitive coupling and charge-division reasoning under the stated model and stimulus.

8 MOS transistor schematics

Source: `01_miller.sp`. All three INV cells use $W_n/W_p/L = 0.6/1.8/0.25\,\mu$m. The explicit coupling capacitor connects the input and output of X2. It is separate from the intrinsic MOS capacitances.

The drawings use the device and node names of the cited netlists. Equal net labels denote an electrical connection. Unless stated otherwise, NMOS bulk terminals connect to ground and PMOS bulk terminals connect to V_{DD}. The bulk wires are omitted from the three-terminal symbols for readability.

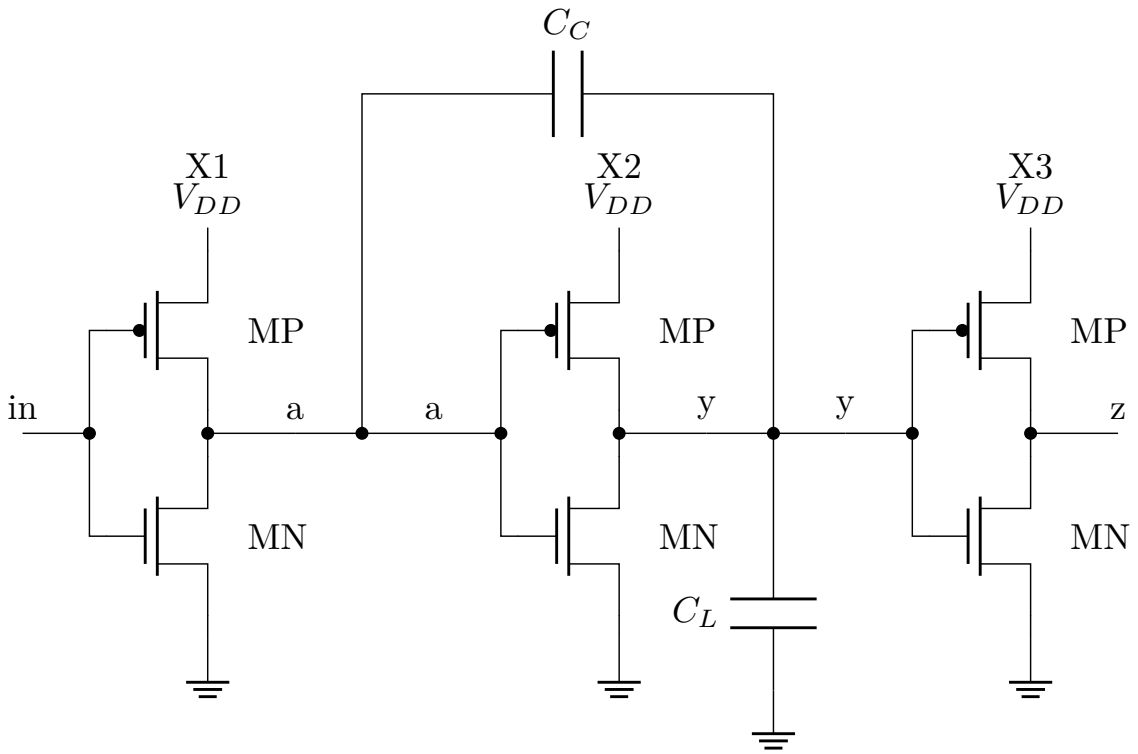


Figure 3: Complete six-MOS signal path. C_C is CCVAL (0/2/5/10 fF); C_L is CLVAL (0/10/40 fF in the selected cases). X3 remains a real gate load when $C_L = 0$. Supply crossings without a dot are not connected.